

CSCE 378: Human Computer Interaction

Lecture: Usability Goals

SEPTEMBER 3RD – ALISHA BEVINS

Today

1. P1 Document
2. Usability Goals
3. User Experience Goals
4. Quiz

Announcements

- P1 due 9/5 by 11:59 pm?

P1 document

3 User Types

Novice → Casual → Expert



Usability Goals

Usability = how easy & pleasant features are to use

Nielsen's Goals:

1. Learnability
2. Efficiency
3. Memorability
4. Error Prevention/ Safety
5. Satisfaction

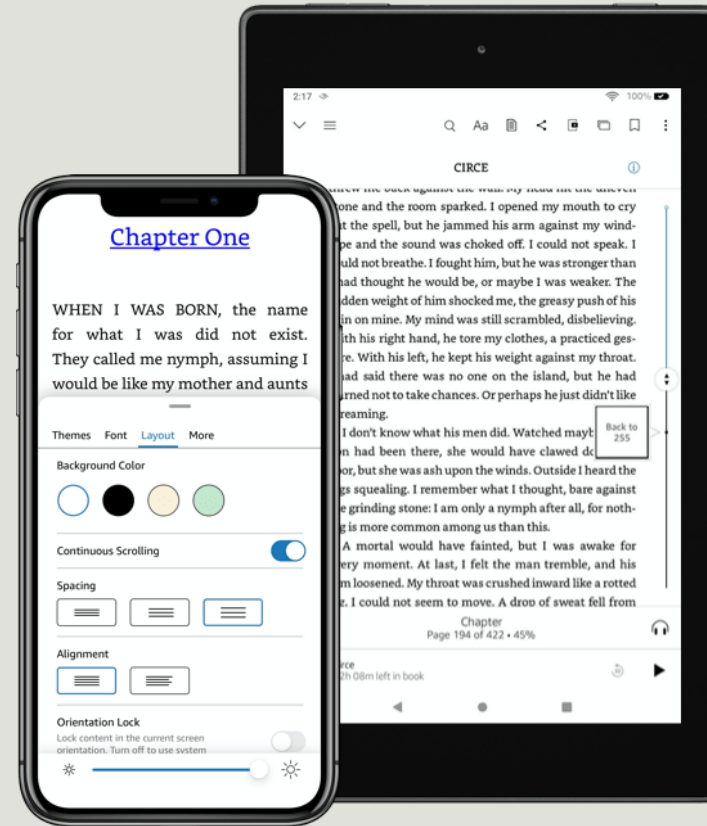
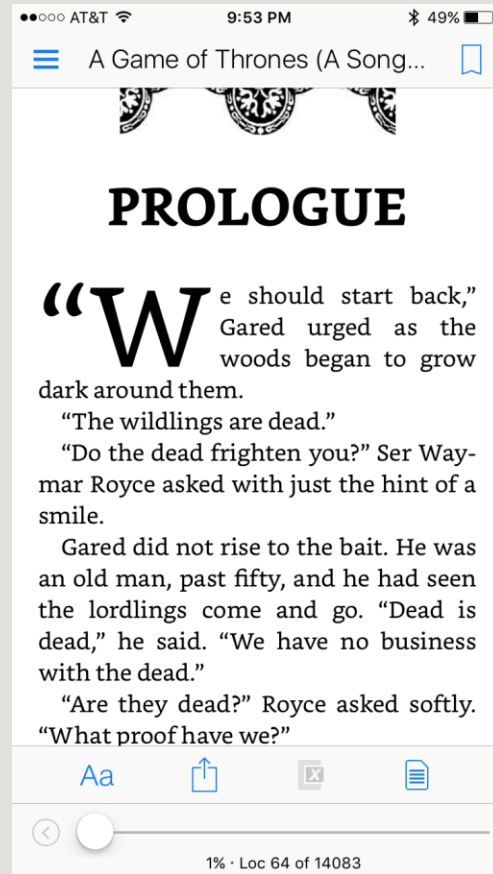
Interaction Design Book (ID):

1. Effectiveness
2. Efficiency
3. Safety
4. Utility
5. Learnability
6. Memorability
7. Satisfaction

Learnability (Definition)

- How easy it is for new users to get started
- Especially important with many novice users
- Key Questions:
 - Can users figure it out by exploring?
 - How hard is it to learn all the functions?

High Learnability



Functions are obvious, discoverable, quick to try

Low Learnability



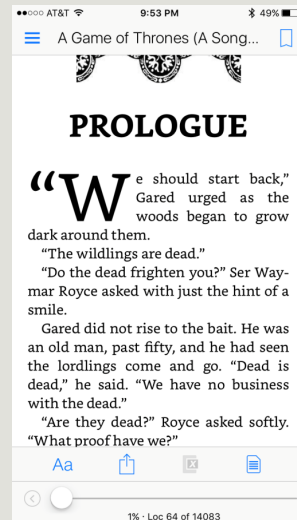
Overwhelming options, requires expert training

Measuring Learnability

Key Test: How long does it take a novice to do basic tasks?

Goals:

- Support learning through exploration
- Make all core functions easy to learn

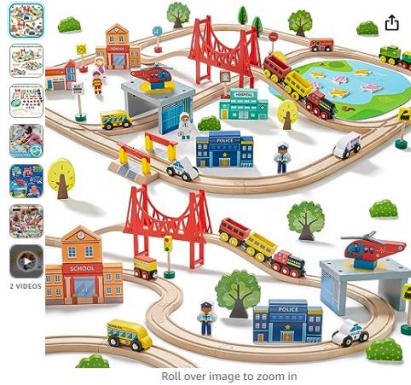


Efficiency

- Productivity once the system is learned
- Most relevant for intermediate & expert users
- Key Questions:
 - How many steps to complete a task?
 - Can stored info reduce effort?

High Efficiency

Toys & Games › Preschool › Pre-Kindergarten Toys › Activity › Trains & Train Sets



Tiny Land Train Set 110pcs Wooden Train Set, Toy Train for Boys & Girls with Wooden Train Track, Wooden Toys for 3-7 Years Old Toddlers & Kids, Railway Set

Visit the Tiny Land Store
4.7 ★★★★★ 3,262 ratings | 35 answered questions
1K+ bought in past month

1 Price Change

\$62⁹⁹
✓prime Two-Day
FREE Returns

Coupon: ✓ \$6 coupon applied to one item per order at checkout [Shop items](#) › | [Terms](#)

Thank you for being a Prime member. Get \$125 off: Pay \$0.00 \$62.99 upon approval for Prime Visa.

Extra Savings Save 10% on White Battery Operated Action Locomotive T... 4 Applicable Promotion(s) ▼

Style: 110PCS Train Set

110PCS Train Set \$62.99 ✓prime	55PCS Crane Train Set \$32.99 ✓prime	Wooden train set --
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Purchase options and add-ons

Payment plans
\$11.44/mo (6 mo) at example APR of 30% (rates from 10-30% APR)

- [110PCS All-in-One Wooden Train Set Perfect for Family & Party Time] This 110-piece set comes with wooden train tracks, scenario building blocks, and puzzles. This large-scale train set is appropriate for children aged 3 to 8, allowing them to experiment with various assembly method and enjoy a range of play options while fostering teamwork for up to three children. It is also the finest option for a fun

Roll over image to zoom in

2 VIDEOS

Delivery Pickup

\$62⁹⁹
✓prime Two-Day
FREE Returns

FREE delivery **Thursday, August 24.** Order within 20 hrs 48 mins

📍 Deliver to Alisha - Lincoln 68503

In Stock

Qty: 1 ▼

Add to Cart

Buy Now

Payment Secure transaction
Ships from Amazon
Sold by Miennes
Returns Eligible for Return, Refund or Replacement within 30 days of receipt
Packaging Shows what's inside
☐ Add a gift receipt for easy returns

Add to List ▼
Add to Baby Registry
Add to Registry & Gifting ▼

Few steps, fast task completion

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Low Efficiency

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Fall 2023

BUILD SCHEDULE & ENROLL

TEXTBOOK ONLINE ORDERING

DROP

VIEW SCHEDULE

SHOPPING CART

COURSE	TITLE	DAY	TIME	LOCATION	CREDITS	INSTRUCTOR	STATUS
CSCE - 378 - 001	HUM-COMPUTR INTERACT	MWF	03:30PM - 04:20PM	LPH-102 CITY	3	Alisha Bevins	O

Search by Class Nbr.

SEARCH

Click [here](#) for classic enroll

VIDEOS

- **Beginner:** Scheduler Planner Basics
- **Intermediate:** Customizing Campus Options
- **Intermediate:** Using Locks to Narrow Down Schedules
- **Intermediate:** Customizing Parts of Term Options
- **Advanced:** Using Custom Course Options

INSTRUCTIONS

Building your Schedule

Search for Classes

Class Details

Add/Drop

Changing Class Options

Many steps, slow task completion

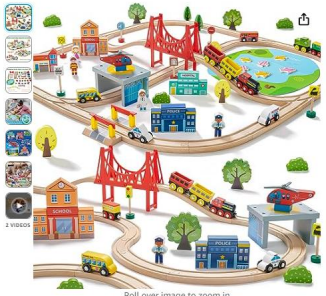
Measuring Efficiency

Key Test: *How much effort per task?*

Metrics:

- Steps/keystrokes
- Pages/screens
- Downtime

Toys & Games • Preschool • Pre-Kindergarten Toys • Activity • Trains & Train Sets



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Fall 2023

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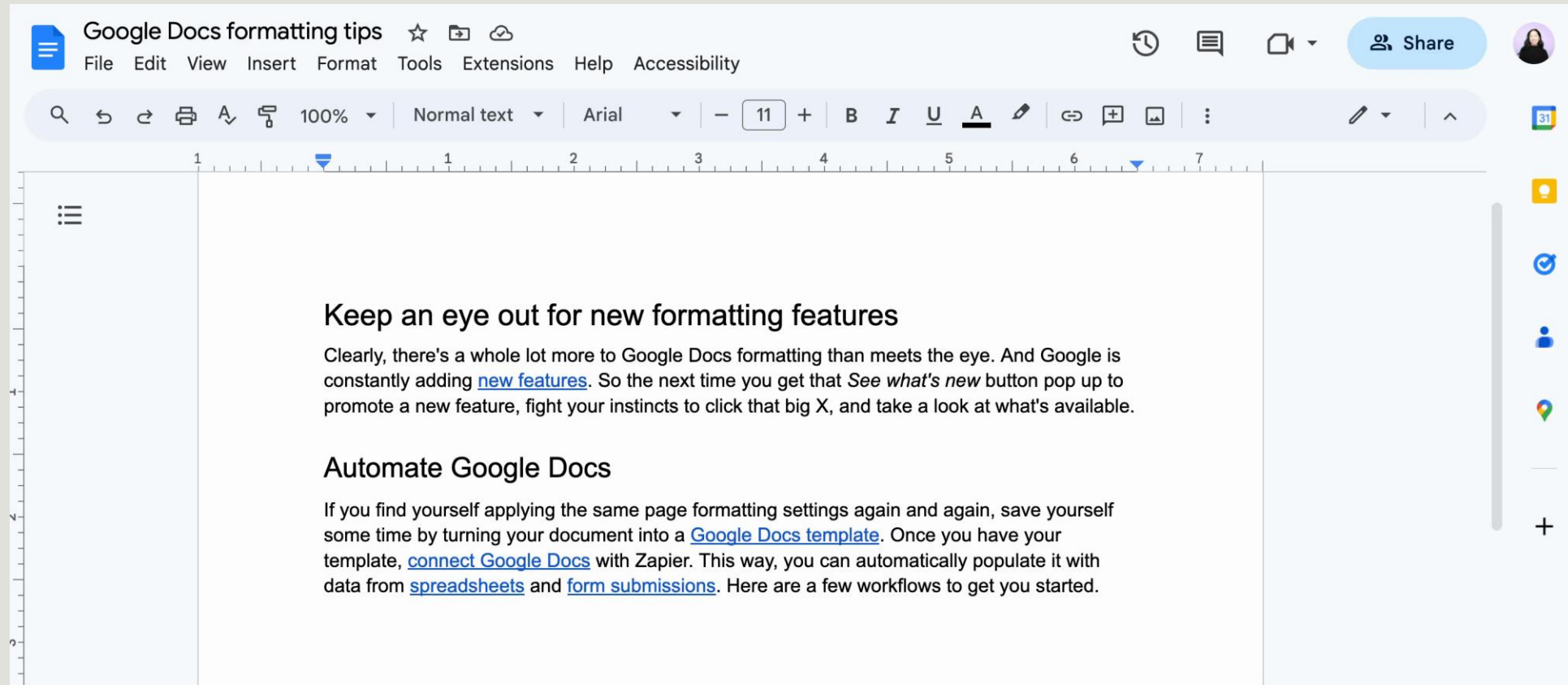
INSTRUCTIONS

- Building your Schedule
- Search for Classes
- Class Details
- Add/Drop
- Changing Class Options

Memorability

- How easy a product is to remember once learned
- Especially important for infrequent tasks
- Key Question: What supports help users remember?

High Memorability



Consistent layouts and icons across products

Low Memorability



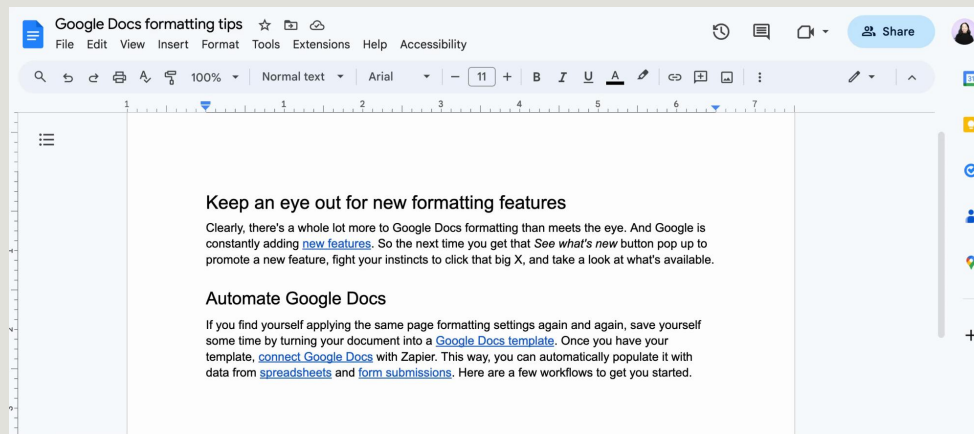
Infrequent use + overloaded controls = confusion

Measuring Memorability

Key Test: *After time away, can users still perform tasks?*

Goals:

- Support remembering
- Support infrequent use



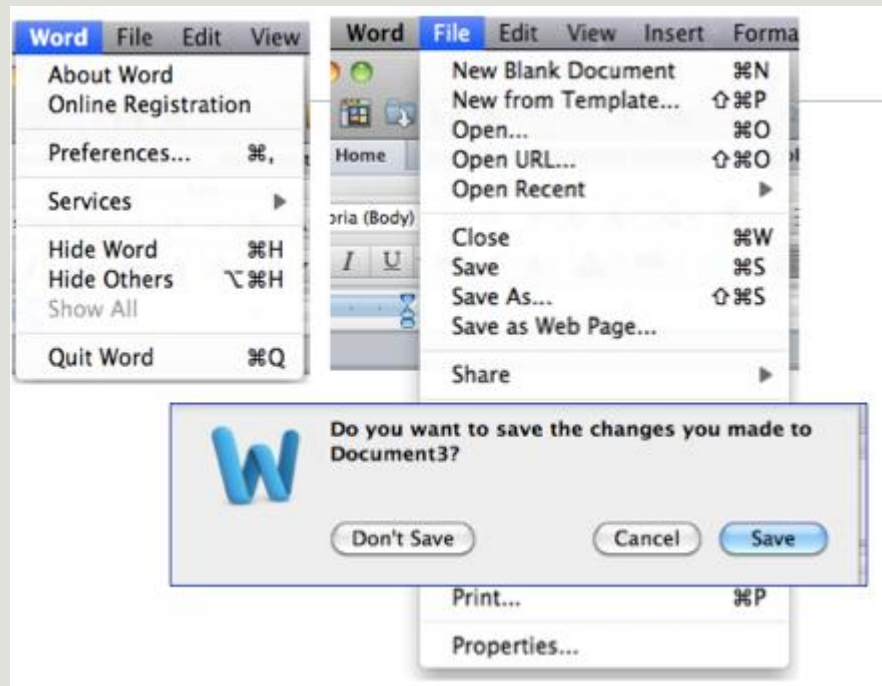
Errors/Safety

Protecting the user from dangerous conditions and undesirable situations, also from perceived fear

Questions:

- What is the range of possible errors?
- What measures to recover easily from them?

Errors (High Safety)



Anticipates common mistakes

Errors (Low Safety)



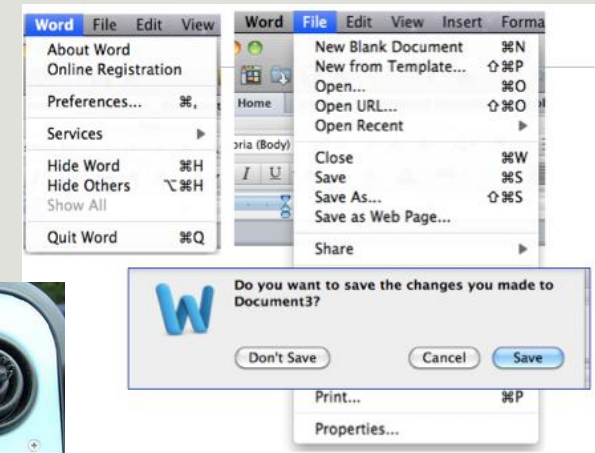
Critical errors are easy to make

Measuring Errors/Safety

Key Test: How often do users make mistakes or lose data?

When most important:

- Frequent errors
- Lost data
- High consequences



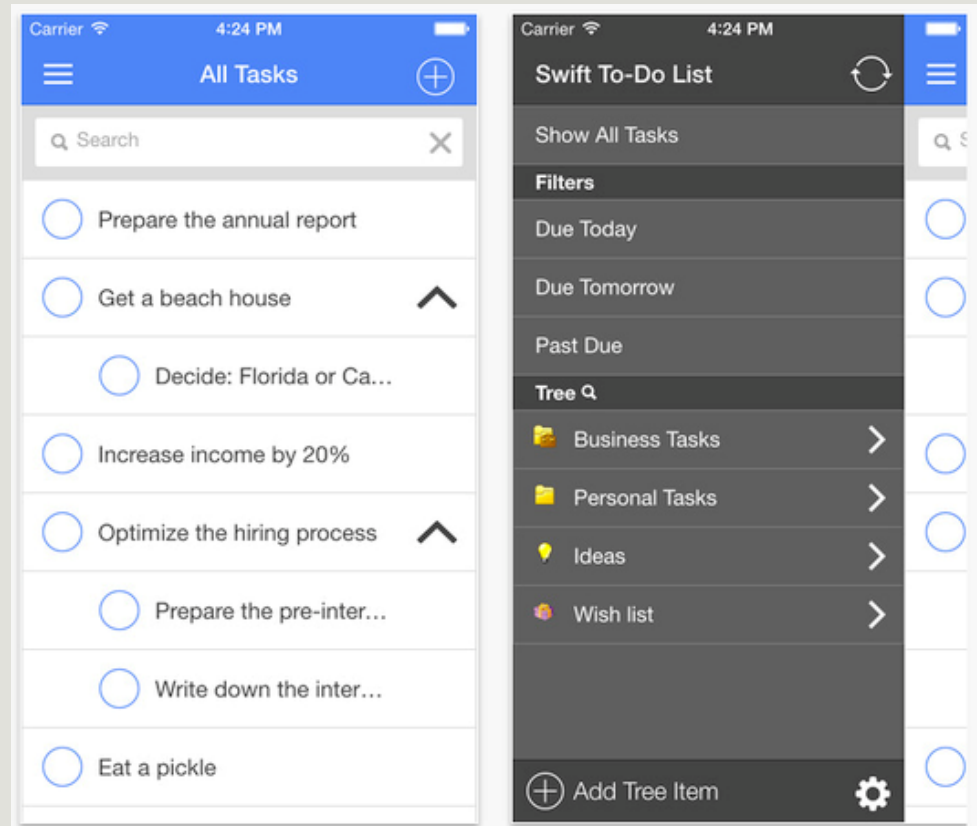
Satisfaction

Subjective quality of how a system feels

Key Question:

- What's the user's response after completing a task?

High Satisfaction

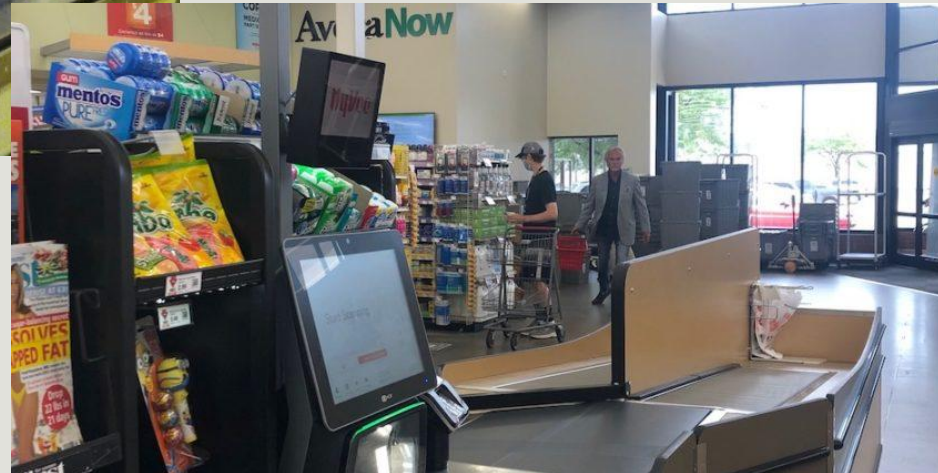


Satisfying interactions encourage repeat use

Low Satisfaction



Frustration outweighs success of task



Measuring Satisfaction

- Do users want to use it again?
- Compare against alternatives
- Most important when users have a choice

EST. 1942

Hem & Stitch
CLOTHING CO.

How likely are you to recommend
Hem & Stitch to a friend?

0

1

2

3

4

5

6

7

8

9

10

Not likely Very likely

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Usability Goals Recap

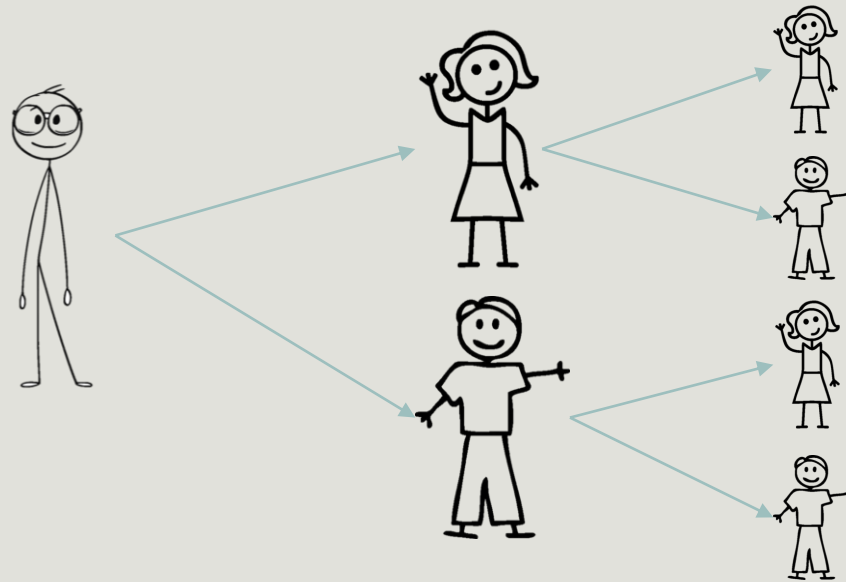
- Learnability
- Efficiency
- Memorability
- Safety/Errors
- Satisfaction

Which goals matter most depends on your context.

Usability in Business

10% of budget should go towards design

Why is usability important?

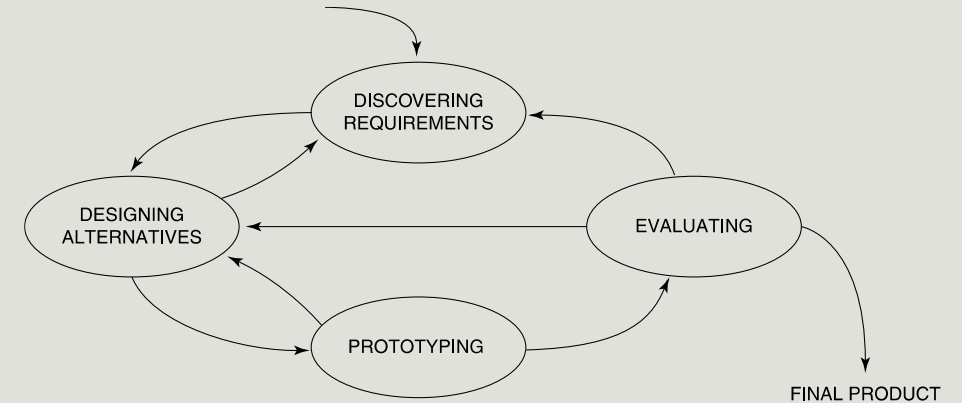


User Experience Goals

Desirable aspects		
Satisfying	Helpful	Fun
Enjoyable	Motivating	Provocative
Engaging	Challenging	Surprising
Pleasurable	Enhancing sociability	Rewarding
Exciting	Supporting creativity	Emotionally fulfilling
Entertaining	Cognitively stimulating	Experiencing flow
Undesirable aspects		
Boring	Unpleasant	Creepy
Frustrating	Patronizing	Intrusive
Making one feel guilty	Making one feel stupid	Invasive
Annoying	Cutesy	Deceptive
Childish	Gimmicky	Annoying

Key Points

- Four basic activities in interaction design process
 - Discovering requirements
 - Designing alternatives
 - Prototyping
 - Evaluating
- People-centered design rests on three principles
 - Early focus on users and tasks
 - Empirical measurement
 - Iterative design
- Understand problem space before starting design
- Involve stakeholder groups throughout development
- Generate alternative designs and evaluate them



Quiz on Canvas
